

Customer Journey Map

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	2 Marks

Customer Journey Map

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

Phase of Journey	Stage 1 Soil Sampling & Input Configuration	Stage 2 Engine Inference & Optimization	Stage 3 Precision Sowing & Yield Execution
Actions <i>What does the customer do?</i>	• Extracts uniform topsoil core samples across the designated production fields. • Acquires chemical N-P-K ratios and pH levels via an agile field test kit or local lab report. • Launches the OptiCrop web platform and prepares the multi-dimensional parameter matrix form.	• Inputs specific soil configuration vectors alongside localized historical ambient rainfall numbers. • Executes the predictive engine model using the central interactive recommendation trigger. • Compares recommended secondary crop compatibility indexes for fallback options.	• Procures optimized high-yield seed stocks matching the system's absolute primary recommendation. • Applies precise calibration rules for targeted nitrogen/potassium fertilizer usage based on data insights. • Monitors physical growth outputs and logs production milestones back into the analysis modules.
Touchpoint <i>What part of the service do they interact with?</i>	• OptiCrop Responsive Web Application Landing View. • Configured Data Extraction Form Inputs. • Physical Soil Matrix Data Logs.	• Flask Backend Application Processing Routers. • Dynamic Machine Learning Prediction Panel. • Visual Compatibility & Constraint Threshold Chart Displays.	• Physical Harvest Fields and Seed Distribution Networks. • Optimization Diagnostics Report Output. • Platform Yield Feedback & Regional Researcher Portals.

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Customer Thought <i>What is the customer thinking?</i>	• "Are my raw NPK field inputs precise enough to feed the predictive ML classification models?" • "I hope this interface is straightforward to navigate on a standard smartphone screen."	• "The matching between this crop recommendation and my field's historic performance looks highly accurate." • "Can I easily adapt my preset operational budgets to fulfill these recommended changes?"	• "My structural operating costs decreased because I eliminated unnecessary chemical fertilizer purchases." • "This data-driven approach removes all the stressful guesswork from my seasonal sowing plan."
Customer Feeling <i>What is the customer feeling?</i>	• Anxious yet eager regarding the digital parameters configuration. • Uncertain about moving away from classic traditional intuitive cultivation habits.	• Enlightened by seeing calculated nutrient compatibility levels clearly visualized. • Reassured by seeing localized metrics match up with advanced model projections.	• Empowered by achieving superior yields with minimized material resources. • Confident in securing predictable income streams through precision agriculture engines.